

Yinquan WANG

DATA SCIENTIST · REINFORCEMENT LEARNING · OPERATIONS RESEARCH · AGENTIC AI

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PROFILE

Data scientist and researcher at the intersection of reinforcement learning, operations research, and agentic AI. Experienced in translating research prototypes into deployable decision systems for transportation, manufacturing, district heating, industrial analytics, and tool-using agents.

EXPERIENCE

Data Scientist · R&D; Department, Beijing Medical Health Large Model Co., Ltd. Jul 2024 - Present

Industrial internet, operational optimization, deep learning, large language models, and data-science agents.

- Design scheduling, forecasting, control, and anomaly-detection solutions for operational industrial systems.
- Build multi-user agent workflows with file/context management, tool execution, approval recovery, and artifact rendering.

Assistant Engineer · Traffic Information Center, Beijing Transportation Commission Mar - Jun 2019

- Contributed to the Specifications for Labeling Text Information of Transportation Video Images.

SELECTED IMPACT

Ride-sourcing dispatch

RL dispatch methods improved response rate by 4.15% and platform revenue by 7.30%.

Industrial scheduling

Deployed scheduling systems across factories; reduced development time from 3 months to 1 and improved steel utilization by 1%.

District heating

Covered 130+ heat-exchange stations and 8,000+ industrial data points, achieving at least 10% electricity savings.

Industrial anomaly detection

Reduced gas-anomaly false alarms by 25%; expanded to 33+ mines and 340+ working faces.

EDUCATION

Ph.D. in System Science · Beijing Jiaotong University 2019 - 2024

Dissertation: Passenger-driver matching and vehicle dispatch algorithms in ride-sourcing markets.

Visiting Ph.D. Researcher · Hiroshima University 2022 - 2023

M.S. in Transportation Planning and Management · Beijing Jiaotong University 2018 - 2019

Continued in the combined master's and doctoral program.

B.Eng. in Traffic Engineering · Qingdao University of Technology 2014 - 2018

AGENTIC AI & OPTIMIZATION
PROJECTS

Data science agent platform

Multi-user project workspaces, model/provider configuration, file upload, artifact rendering, and approval recovery for long-running tool calls.

Business data QA agent

A single-agent chain for business data query, knowledge QA, artifact generation, and platform-function orchestration.

OR-Agent / solver agent

Translate natural-language tasks into mathematical models, call open-source or commercial solvers, inspect outputs, and learn from verifier feedback.

RL post-training environments

Build CodeAct trajectories, executable tools, verifiable rewards, and RLVR-oriented training data.

SELECTED PUBLICATIONS

- Y. Wang et al. Promoting Collaborative Dispatching in the Ride-Sourcing Market With a Third-Party Integrator. IEEE T-ITS, 2024. DOI: 10.1109/TITS.2023.3348764
- Y. Wang et al. Reinforcement Learning-Based Order-Dispatching Optimization in Ride-Sourcing Service. Computers & Industrial Engineering, 2024. DOI: 10.1016/j.cie.2024.110221
- Y. Wang et al. Reassignment Algorithm of the Ride-Sourcing Market Based on Reinforcement Learning. IEEE T-ITS, 2023. DOI: 10.1109/TITS.2023.3274636
- Y. Wang et al. Order Dispatching Optimization in Ride-Sourcing Market by Considering Cross Service Modes. Journal of Central South University, 2023. DOI: 10.1007/s11771-022-5193-4

TECHNICAL SKILLS

RL & deep learning	PyTorch, Gym, Stable-Baselines3, simulation, PPO, GRPO, RLVR
Operations research	CPLEX, DOcplex, mathematical programming, scheduling, matching, routing, learning-augmented optimization
Data science	Pandas, NumPy, SciPy, scikit-learn, GeoPandas, Spark, Hadoop, time-series forecasting
Agent systems	CodeAct, ReAct, tool integration, verifiers, SFT, RAG, vLLM, human-in-the-loop approval
Engineering	Python, SQL, Django, DRF, JavaScript, React, MySQL, ECharts, Leaflet, Mapbox GL

HONORS

- 2025 Striver of the Year, China Unicom Data Intelligence Co., Ltd.
- National Second Prize, Digital China Innovation Competition, 2020.
- National Third Prize, Mathematical Contest in Modeling for Chinese Graduate Students, 2019.
- Beijing Jiaotong University Doctoral Scholarship, 2019-2021.